

Chapter 1

An Introduction to Macroeconomics

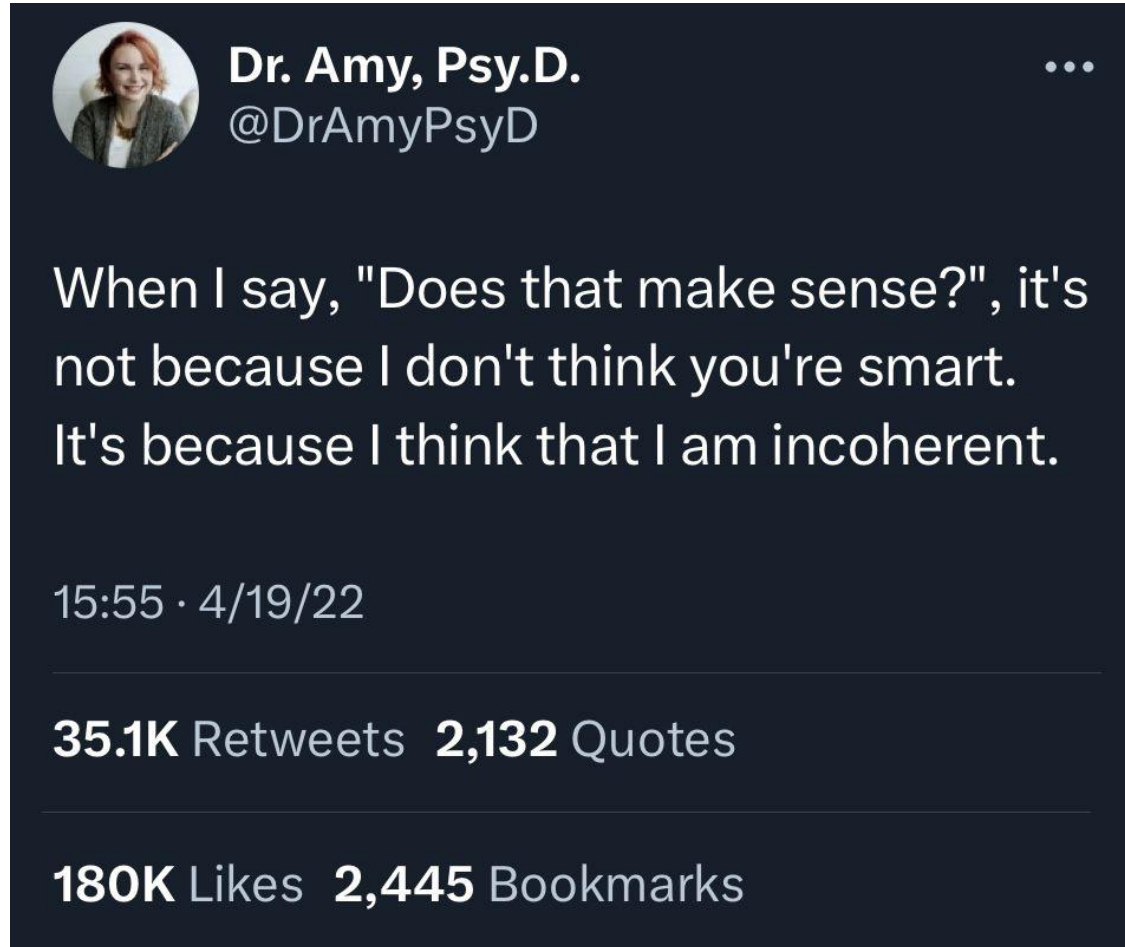
Dr. Matt D'Urso

(Colgate University)

Housekeeping

- › © [2026] [Dr. Matt D'Urso]. All rights reserved.
- › The compilation, structure, original text, and design of these lecture slides are the intellectual property of [Matt D'Urso]. All third-party figures, images, quotes, and cited research remain the property of their respective copyright holders and are used here under fair use guidelines for non-commercial educational purposes.
- › No part of these presentations may be reproduced, modified, republished, distributed, or converted into commercial works (including textbooks, online courses, or study materials for sale) without express written permission from the author.
- › **Free for personal educational use.** *Downloads are for individual study only. Re-hosting, commercial distribution, or derivative works (such as turning these into a textbook or course package) are strictly prohibited without written consent.*

Before we begin...



(Marschall, 2022). When I say cite everything, I mean it!!!!

What is Economics?

- › **Economics** is the study of the allocation of scarce resources and their production, distribution, and consumption

What is Economics, Really?

- › Economics is *really* the study of human decision making and how those decisions influence resource allocation

- › What are these decisions?
 - They are essentially microeconomics
 - Modern macro is “micro-founded”
 - It is built on microeconomics

What Does This Mean for Macro?

- › In general, we will study how micro-level decision making moves aggregate variables and prices
- › National output does not make decisions; people do
- › We study how these decisions aggregate to influence things like total output, investment, consumption, etc.
 - Aggregate simply means “to add up”

Does This Matter?

- › If we want to study aggregate outcomes, why not just look at aggregates directly?
 - Just look at total output before/after an event
- › We can, but this only provides us with a surface level view
 - Can't see what's going on "under the hood"
- › We can only speculate about what is causing any changes

What do Microfoundations Tell Us?

- › People make decisions to maximize their expected utility
 - Translation: people try their best, given any constraints, to make the choice they think will make them happiest
- › Possible constraints:
 - Budget, time, space, energy, available information
 - › Smoking rates fell when more health information was available
- › These decisions are also **dynamic**; they occur over time
 - Make choices today, based on expected budget tomorrow

Dynamic Decisions: The Permanent Income Hypothesis

- › People don't spend out of their "current" budget, they spend out of their "Permanent" or "Lifetime" budget
 - People prefer to "smooth" large costs over time
 - Small hits to monthly lifetime budgets vs large hit to current budget
 - › E.g. borrowing to go to college or purchase a house
 - Friedman, Milton. "A Theory of the Consumption Function." Princeton University Press. 1957.

- › Similar research on "Life-cycle" work
 - Modigliani, Franco, and Richard Brumberg. "Utility Analysis and the Consumption Function." 1954.

Research and Policy Implications

- › This has huge consequences for research and policy
- › If we want our models to accurately depict what we're trying to study, we need to be mindful of proper microfoundations
 - Something to think about: is it enough for a model to just make accurate predictions, or should it be accurate to the real world?
- › If we want to conduct policy that helps people, we need to be mindful of how they will make decisions

The Lucas Critique

› What we know:

- People try to make the best choices given constraints
- These choices can change when the constraints change
- These decisions occur over time

› *If policy changes someone's constraints, then we should expect their decisions to change. Moreover, we should be aware of this when forming policy in the first place.*

- Lucas Jr, Robert E. "Econometric policy evaluation: A critique." Carnegie-Rochester conference series on public policy. Vol. 1. North-Holland, 1976.

An Example with Taxes

- › A country is increasing consumption taxes to raise revenue
 - People expect taxes to go up tomorrow
 - Response: consume more today to avoid higher costs tomorrow
- › With less consumption tomorrow, there is less tax revenue
 - Policy can respond by raising taxes even more to make up the difference...
- › But individuals' decisions will respond again

Analysis Without Dynamic Responses

- › What would happen if we didn't consider the response to this policy?
- › If previous data says a 1% increase in taxes leads to a 4% increase in revenue, should a 2% increase in taxes lead to an 8% increase?
 - Why or why not?
 - We're looking at percent change, not the level of taxes. Does percent change account for the dynamics we're talking about?

A Strict Example: Ricardian Equivalence

- › Consider a government trying to boost the economy through spending
- › If people expect an increase in taxes in the future to pay for this, they will spend less now to save for the taxes coming in the future
- › Less household spending offsets government spending. The case where they cancel out is Ricardian Equivalence
 - Ricardo, David. "Essays on the Funding System." 1820.

Microfoundations Matter

- › Is Ricardian Equivalence strictly true? Probably not
 - But most macroeconomists argue for is the importance of being consistent with micro-behavior
- › Model building will have some type of equilibrium behavior for individuals (competitive, monopolistic, oligopolistic, etc.)
- › Then aggregate results and compare to real-world data

The [quick] History of Macroeconomics

- › In an ironically recursive way, we can look at the study of macroeconomics as a series of dynamic decisions
- › We update our forecast of the field, which updates our forecasts of the macroeconomy, which updates our forecasts of the field...

How We Got Here: Classical Economics

- › 1776: Adam Smith – “The Invisible Hand”
 - Smith, Adam. “An Inquiry into the Nature and Causes of the Wealth of Nations.” (1776).
- › 1817: David Ricardo – comparative advantage vs mercantilism
 - Ricardo, David. “On the Principles of Political Economy and Taxation.” (1817).
- › 1870’s: Carl Menger – marginalism and subjective theory of value
 - See also Walras, Marshall, Jevons (economics is “mathematical”)
 - Menger, Carl. “Principles of Economics.” (1871).
 - Jevons, William S. “A General Mathematical Theory of Political Economy. (1862).
 - Marshall, Alfred. “Principles of Economics.” (1890).
- › 1874: Léon Walras – General Equilibrium
 - Walras, Léon. “Elements of Pure Economics.” (1874).

How We Got Here: Keynesian Revolution

- › 1936: John M. Keynes – “The General Theory”
 - Written during Great Depression
 - Keynes, John M. “The General Theory of Employment, Interest and Money.” (1936).
- › Prices are **sticky**; they take time to adjust
 - Wage stickiness prolongs unemployment
- › Government intervention can help:
 - Raise taxes/cut spending in good times to build funds
 - Lower taxes/raise spending in bad times to help unemployment

How We Got Here: Neoclassical Synthesis

- › WWII – 1970: Consensus that in the short-run, Keynesian analysis was correct. In the long-run, markets equilibrate
 - John Hicks and Alvin Hansen: IS-LM
 - › Hicks, John R. "Mr. Keynes and the "classics"; a suggested interpretation." *Econometrica: journal of the Econometric Society* (1937): 147-159.
 - › Hansen, Alvin. "Monetary Theory and Fiscal Policy." (1947).
 - Paul A. Samuelson: Modern basis for mathematical economics
 - › Samuelson, Paul A. "Foundations of Economic Analysis." *Harvard University Press* (1947).
 - Robert Solow: Long-run growth
 - › Solow, Robert M. "A contribution to the theory of economic growth." *The quarterly journal of economics* 70.1 (1956): 65-94.

How We Got Here: Monetarism

- › 1960 – 1990: “Inflation is always and everywhere a monetary phenomenon”
 - Milton Friedman and Anna Schwartz
 - › Friedman, Milton, and Anna Jacobson Schwartz. "A Monetary History of the United States, 1867-1960." (1963).
 - › Friedman, Milton. "A Theoretical Framework for Monetary Analysis." *Journal of Political Economy* 78.2 (1970): 193-238.
- › Revival of the Quantity Theory of Money and impact on inflation and expectations of inflation
 - Something Keynes thought of as second-order
 - Very popular during high inflation episodes of the 1970s

Today: Real Business Cycle Theory

- › 1976 – today: Focus on real frictions causing changes to **relative prices**
 - Price of one good in terms of another
 - Assumes people view trade-offs in that sense
 - Rational expectations about future values

- › Began with Lucas Critique; driven by Kydland and Prescott
 - Kydland, Finn E., and Edward C. Prescott. "Time to build and aggregate fluctuations." *Econometrica: Journal of the Econometric Society* (1982): 1345-1370.
 - Prescott, Edward C. "Theory ahead of business-cycle measurement." *Carnegie-Rochester conference series on public policy*. Vol. 25. North-Holland, 1986.
 - King, Robert G., Charles I. Plosser, and Sergio T. Rebelo. "Production, growth and business cycles: I. The basic neoclassical model." *Journal of monetary Economics* 21.2-3 (1988): 195-232.

Today: New Keynesian Economics

› 1970s – today: Merging expectations with price stickiness

- Fischer, Stanley. "Long-term contracts, rational expectations, and the optimal money supply rule." *Journal of political economy* 85.1 (1977): 191-205.
- Rotemberg, Julio J. "Monopolistic price adjustment and aggregate output." *The Review of Economic Studies* 49.4 (1982): 517-531.
- Calvo, Guillermo A. "Staggered prices in a utility-maximizing framework." *Journal of monetary Economics* 12.3 (1983): 383-398.

› Elements of imperfect competition and coordination failures of the market

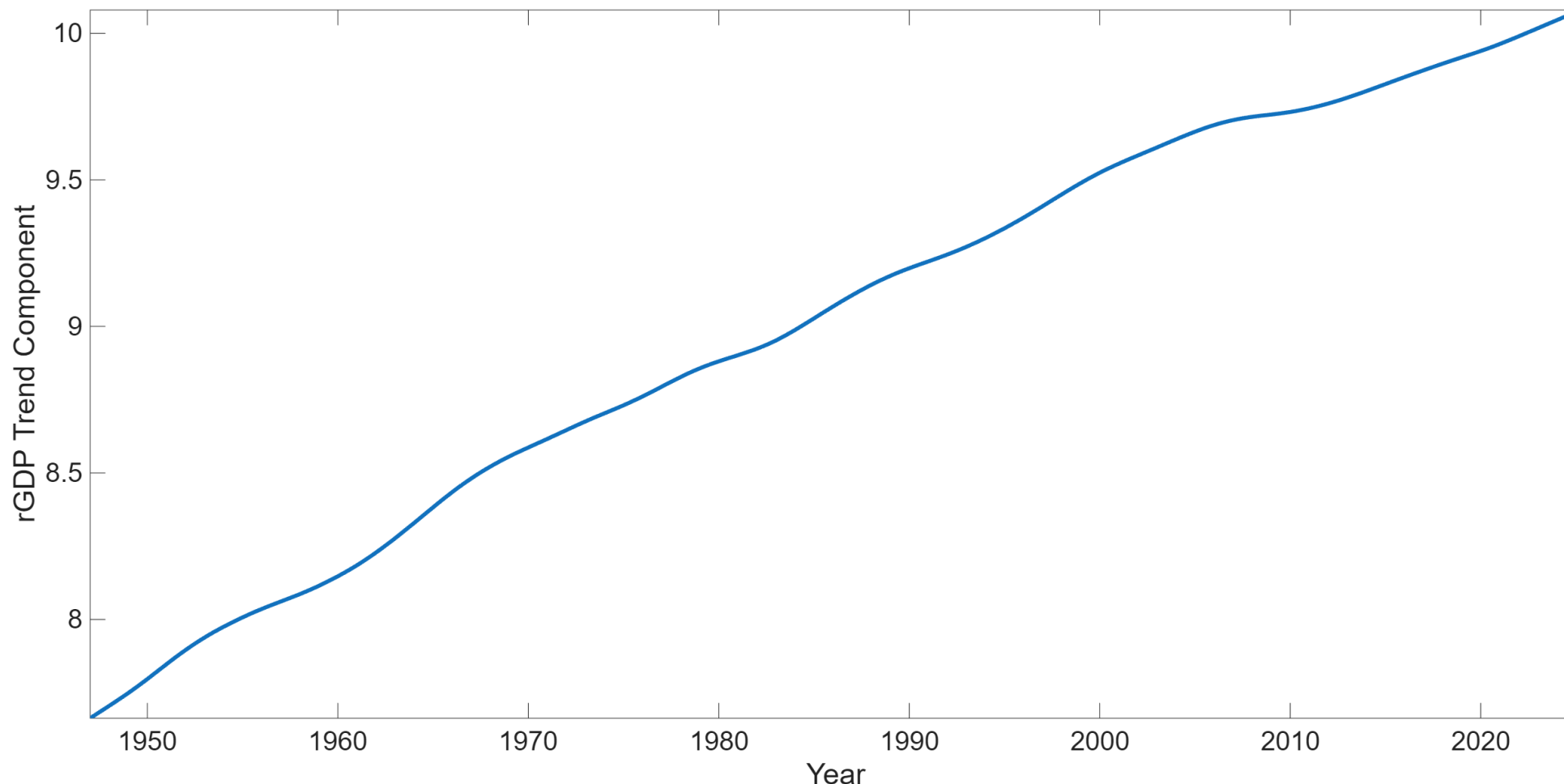
- Diamond, Peter A. "Aggregate demand management in search equilibrium." *Journal of political Economy* 90.5 (1982): 881-894.

› **New Neoclassical Synthesis:** Real Business Cycle Theory + New Keynesian Economics

- What we'll study this semester

π

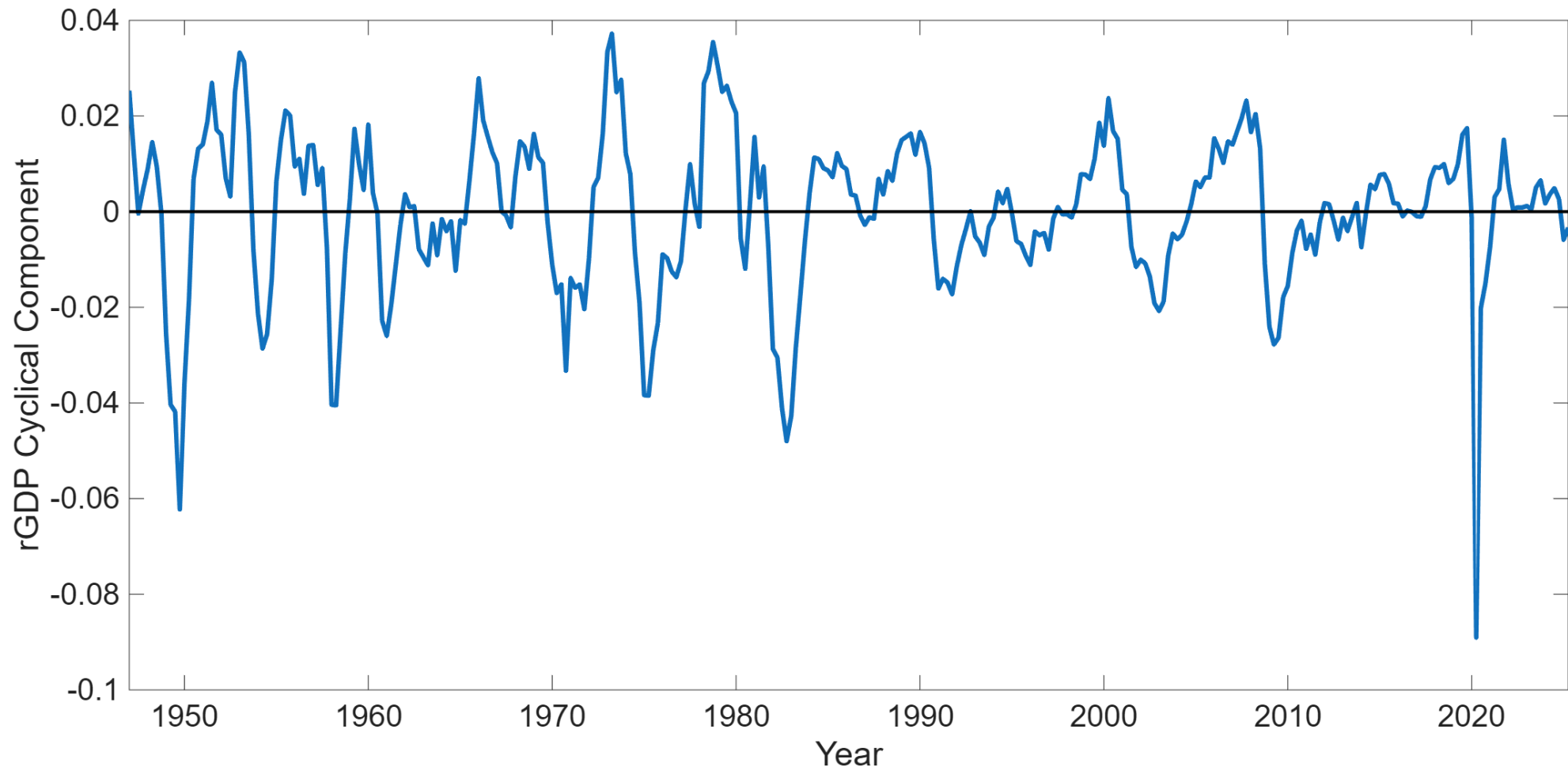
Today: In the Long Run, Markets Clear,...



HP filtered data of log of quarterly U.S. real GDP (chained 2017 dollars, seasonally adjusted annual rate);
 $\lambda=1600$. Data: Real Gross Domestic Product (BEA, 2026)

π

...but in the Short Run, Frictions Matter



HP filtered data of log of quarterly U.S. real GDP (chained 2017 dollars, seasonally adjusted annual rate);
 $\lambda=1600$. Data: Real Gross Domestic Product (BEA, 2026)

Alternative Approaches (not covered)

- › Socialist economics:
 - Abba Lerner
 - Oscar Lange
- › Laissez-faire:
 - Friedrich Hayek
 - Ludwig von Mises
- › Debates about Economic Calculation:
 - Do markets accurately tell us how people value resources?
 - If not, what can?

Beyond the Stock Market and Production

- › Four decades of topics not traditionally thought of as Economics:
- › Reparations and Racial Wealth Dynasties
 - Boerma, Job, and Loukas Karabarbounis. "Reparations and persistent racial wealth gaps." *NBER Macroeconomics Annual* 37.1 (2023): 171-221.
- › Traffic congestion's impact on birth weights (*fan favorite*)
 - Currie, Janet, and Reed Walker. "Traffic congestion and infant health: Evidence from E-ZPass." *American Economic Journal: Applied Economics* 3.1 (2011): 65-90.
- › Moneyball, viewed through economics
 - Hakes, Jahn K., and Raymond D. Sauer. "An economic evaluation of the Moneyball hypothesis." *Journal of Economic Perspectives* 20.3 (2006): 173-185.
- › Experiments on how people perceive fairness under uncertainty
 - Kagel, John H., Chung Kim, and Donald Moser. "Fairness in ultimatum games with asymmetric information and asymmetric payoffs." *Games and Economic Behavior* 13.1 (1996): 100-110.

References (1/3)

1. Boerma, Job, and Loukas Karabarbounis. "Reparations and persistent racial wealth gaps." *NBER Macroeconomics Annual* 37.1 (2023): 171-221.
2. Calvo, Guillermo A. "Staggered prices in a utility-maximizing framework." *Journal of monetary Economics* 12.3 (1983): 383-398.
3. Currie, Janet, and Reed Walker. "Traffic congestion and infant health: Evidence from E-ZPass." *American Economic Journal: Applied Economics* 3.1 (2011): 65-90.
4. Diamond, Peter A. "Aggregate demand management in search equilibrium." *Journal of political Economy* 90.5 (1982): 881-894.
5. Fischer, Stanley. "Long-term contracts, rational expectations, and the optimal money supply rule." *Journal of political economy* 85.1 (1977): 191-205.
6. Friedman, Milton, and Anna Jacobson Schwartz. "A Monetary History of the United States, 1867-1960." (1963).
7. Friedman, Milton. "A Theory of the Consumption Function." *Princeton University Press*. 1957.
8. Friedman, Milton. "A Theoretical Framework for Monetary Analysis." *Journal of Political Economy* 78.2 (1970): 193-238.
9. Hakes, Jahn K., and Raymond D. Sauer. "An economic evaluation of the Moneyball hypothesis." *Journal of Economic Perspectives* 20.3 (2006): 173-185.
10. Hicks, John R. "Mr. Keynes and the "classics"; a suggested interpretation." *Econometrica: journal of the Econometric Society* (1937): 147-159.
11. Hansen, Alvin. "Monetary Theory and Fiscal Policy." (1947).
12. Jevons, William S. "A General Mathematical Theory of Political Economy. (1862).
13. Kagel, John H., Chung Kim, and Donald Moser. "Fairness in ultimatum games with asymmetric information and asymmetric payoffs." *Games and Economic Behavior* 13.1 (1996): 100-110.

References (2/3)

14. Keynes, John M. "The General Theory of Employment, Interest and Money." (1936) .
15. King, Robert G., Charles I. Plosser, and Sergio T. Rebelo. "Production, growth and business cycles: I. The basic neoclassical model." *Journal of monetary Economics* 21.2-3 (1988): 195-232.
16. Kydland, Finn E., and Edward C. Prescott. "Time to build and aggregate fluctuations." *Econometrica: Journal of the Econometric Society* (1982): 1345-1370.
17. Lucas Jr, Robert E. "Econometric policy evaluation: A critique." *Carnegie-Rochester conference series on public policy*. Vol. 1. North-Holland, 1976.
18. Marshall, Alfred. "Principles of Economics." (1890).
19. Marschall, Amy, Psy.D. [@DrAmyPsyD]. "When I say, "Does that make sense?", it's not because I don't think you're smart. It's because I think that I am incoherent." Twitter, 19 April 2022, 15:55, <https://x.com/DrAmyPsyD/status/1516505722313396227>.
20. Menger, Carl. "Principles of Economics." (1871).
21. Modigliani, Franco, and Richard Brumberg. "Utility Analysis and the Consumption Function." 1954.
22. Prescott, Edward C. "Theory ahead of business-cycle measurement." *Carnegie-Rochester conference series on public policy*. Vol. 25. North-Holland, 1986.
23. Ricardo, David. "Essays on the Funding System." 1820.
24. Ricardo, David. "On the Principles of Political Economy and Taxation." (1817).

References (3/3)

25. Rotemberg, Julio J. "Monopolistic price adjustment and aggregate output." *The Review of Economic Studies* 49.4 (1982): 517-531.
26. Samuelson, Paul A. "Foundations of Economic Analysis." Harvard University Press (1947).
27. Smith, Adam. "An Inquiry into the Nature and Causes of the Wealth of Nations." (1776).
28. Solow, Robert M. "A contribution to the theory of economic growth." *The quarterly journal of economics* 70.1 (1956): 65-94.
29. U.S. Bureau of Economic Analysis, Real Gross Domestic Product [GDPC1], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/GDPC1>, February 12, 2026.
30. Walras, Léon. "Elements of Pure Economics." (1874).